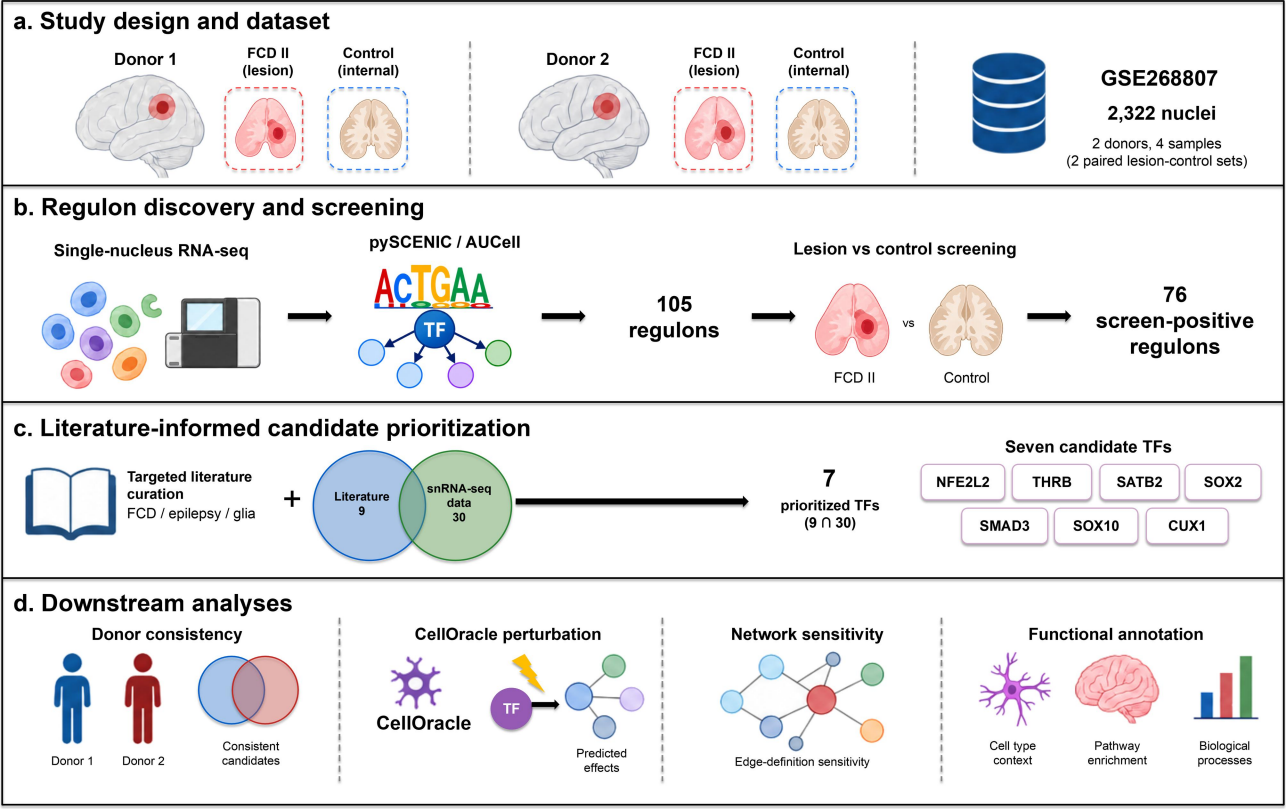

a. Study design and dataset
Donor 1
FCD II (lesion)
Control (internal)
Donor 2
FCD II (lesion)
Control (internal)
GSE268807
2,322 nuclei
2 donors, 4 samples
(2 paired lesion-control sets)
b. Regulon discovery and screening
Single-nucleus RNA-seq
pySCENIC / AUCell
TF
105 regulons
Lesion vs control screening
vs
FCD II
Control
76 screen-positive regulons
c. Literature-informed candidate prioritization
Targeted literature curation
FCD / epilepsy / glia
+
Literature 9
snRNA-seq data 30
7 prioritized TFs (9 ∩ 30)
Seven candidate TFs
NFE2L2
THRB
SATB2
SOX2
SMAD3
SOX10
CUX1
d. Downstream analyses
Donor consistency
Donor 1
Donor 2
Consistent candidates
CellOracle perturbation
CellOracle
TF
Predicted effects
Network sensitivity
Edge-definition sensitivity
Functional annotation
Cell type context
Pathway enrichment
Biological processes